Gaussian Focal Loss: Learning Distribution Polarized Angle Prediction for Rotated Object Detection in Aerial Images

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***Abstract*— With the increasing availability of aerial data, object detection in aerial images has aroused more and more attention in remote sensing community. The difficulty lies in accurately predicting the angular information for each target when using the oriented bounding boxes (OBBs) to represent the arbitrary-oriented objects, as the periodicity of the angle could cause inconsistency between target angle values. To resolve the problem, recent works propose to perform angular prediction from a regression problem to a classification task with a circular smooth label (CSL). However, we find that current loss functions applying to binary soft labels need to approximate the soft label values at each position. When summed over all the negative angle categories, these relatively insignificant loss values can overwhelm the target angle category, thus preventing the network from predicting precise angle information. In this article, we propose a novel loss function that acts as a more effective alternative to the classification-based rotated detectors. By constructing the classification loss with adaptive Gaussian attenuation on the negative locations, our training objective can not only avoid discontinuous angle boundaries but also enable the network to obtain more accurate angle predictions with higher responses at peaks. Moreover, an aspect ratio-aware factor was proposed based on our loss function to enhance the robustness of the model for determining the orientation for square-like objects. Extensive experiments on aerial image datasets DOTA, HRSC2016, and UCAS-AOD demonstrated the effectiveness and superior performances of our approaches.**

***Index Terms*— Aerial images, angle classification, object detection, oriented bounding boxes (OBBs).**

I. Introduction

In RECENT years, the emergence of a large number of aerial images [1]–[3] with higher spatial resolution and richer content has promoted the development of modern remote sensing technologies. Object detection in aerial images has gradually become a hot spot in the field of computer vision [4]–[7]. Benefiting from the deep convolutional neural networks (CNNs), object detection in natural images has achieved great success. Numerous advanced detectors [8]–[14] have been proposed to predict a horizontal bounding box (HBB) for each object. However, the horizontal paradigm for aerial images may suffer from several limitations due to the fact that these images are often taken from a bird's-eye view. So, the object can be densely arranged with various orientations. Simply applying HBBs to these objects will introduce a large amount of irrelevant background or cause significant overlap between targets.

To solve the abovementioned problems, many works [15]–[21] adopted oriented bounding boxes (OBBs) to represent the rotated object more compactly. A common practice is to introduce an extra regression branch for the angle prediction. However, due to the periodicity of the angle, the label of regression value will be discontinuous at the boundary, causing the loss value to a sudden increase. This phenomenon will interfere with the network's learning of object angle information, thereby affecting the final performance of the regression-based rotated detectors. To address the inconsistency issue, advanced works [22]–[24] proposed to convert the angle regression problem to a classification task and use circular smooth labels (CSLs) to replace the one-hot labels. The CSL has two important properties: smoothness and periodicity. The first property is required to reduce the penalties between adjacent angle categories, while the second is designed to tackle the periodic sudden changes in angle. Thus, the CSL-based methods for angle prediction can get rid of the boundary discontinuity issue naturally.

For a classification-based rotated object detector, it will predict a confidence score for each angle interval (category) to obtain an angle distribution vector, and the category with the highest confidence score is used as the final angle prediction result. In practice, the classification-based rotated object detection methods still face a lot of limitations, such as high precision angle estimation and square-like problem. First, current loss functions (focal loss (FL) or binary cross-entropy (BCE) loss) for the binary soft label fail to estimate precise angle information for rotated objects. In contrast to standard classification tasks, angle classification suffers from the characteristic that adjacent angles and angles across boundaries have similar physical meanings. So, the angle classifiers must to be resilient to adjacent angle predictions and have similar prediction results for the cross-boundary angles. The existing methods solve these problems by converting one-hot labels into soft labels and extending the loss functions to the continuous form [e.g., FL to continuous form as in (2)]. However, the current loss paradigm needs to fit the soft label values at each angle category, especially when the loss values at

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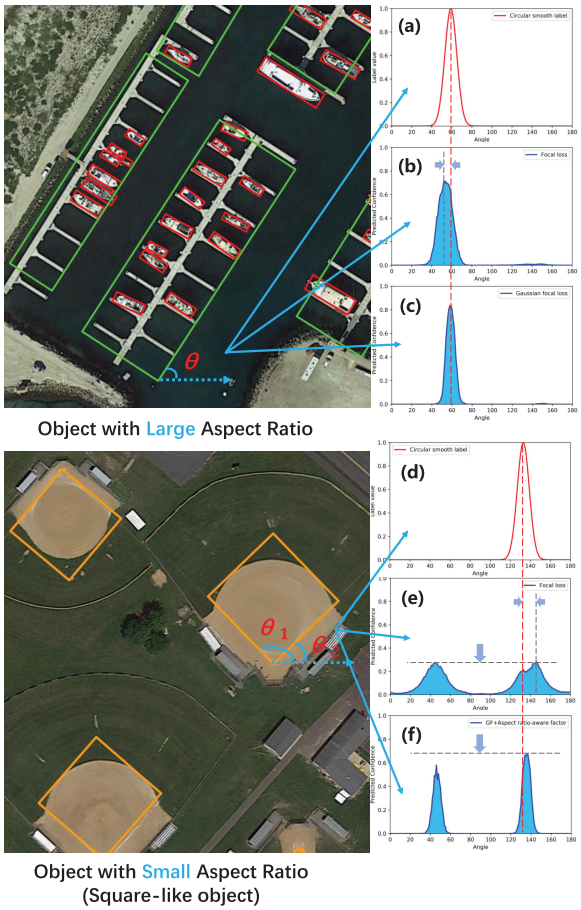


Fig. 1. Visualization of predicted angular distribution from classification-based rotated detectors. (a) and (d) CSLs for some giving samples. The ground-truth angle location is marked with a red dashed line. (b) and (e) Predicted angular distribution from continuous form of FL. (c) and (f) Predicted angular distribution based on our Gaussian FL.

some confusing locations (adjacent angles and cross-boundary angles with target location) are accumulated to override the constraints at the target category. This will cause two effects, first, the accurate prediction of the target angle (the position of angular peak), and second, high confident and reliable evaluation results (the response value at peak). As illustrated in Fig. 1, Fig. 1(a) is the soft angle label (CSL) for a given object, and the target angle position is marked with a red dashed line. Fig. 1(b) is the estimated angular distribution based on continuous FL. We can see that there is still a certain distance between the predicted peak position and the correct angle category, and the classification confidence is not very high. Second, current classification-based rotated detectors may have difficulties in dealing with square-like objects. The combination of "CSL + 180°" can solve the problem of discontinuous boundary naturally but fails for the square-like objects. We find that the ambiguous angle definition of the square-like objects is the main reason for this phenomenon, which we referred as a low confidence and unreliable angle prediction result, as shown in Fig. 1(f).

In this article, we aim to push forward the frontier of classification-based rotated object detectors from the following aspects: First, we propose a Gaussian FL, which adopts a dynamic weighting mechanism to perform precise angle estimation for rotated objects. Compared with the general classification task using a one-hot label, we deal with the influence between adjacent angles by adaptively adjusting the loss weights at locations near the target angle category. Using Gaussian weight decay on the negative locations, the model has a certain elasticity in the prediction of adjacent angles. Meanwhile, due to the circular nature of the decay weights, we can avoid the inconsistency problem caused by the periodicity of the angle. Compared with the angle classification using binary soft labels, we do not need to approximate the label values at each category, thus avoiding a large number of unimportant loss contributions. We use the cross-entropy term to stimulate the response at the target category while suppressing the rest, by which the network can focus more on predicting the precise angle of the object. As shown in Fig. 1, our approach can generate more accurate and high confident angle prediction results [Fig. 1(c)] compared with other [Fig. 1(b)]. Second, we propose an aspect ratio-aware factor to accurately predict the angle information for square-like objects. The five parameters $[c_x, c_y, h, w, \theta]$ is the most extensively applied for the oriented objects (we refer to the long-side definition method [16], [18] unless specified). It does not work for square-like objects, nevertheless, which can be explained by the non-uniqueness of the angle target value when the aspect ratio of an object is close to 1 from our perspective. As shown in the example of Fig. 1 (bottom case), when θ is determined by the long side of the rectangle and x-axis, the network is then ambiguous in discriminating the reference side, causing instability in the training process and eventually producing poor angle prediction results [Fig. 1(e)]. In this context, we design an aspect ratio-aware factor based on our Gaussian FL, where the key idea lies in dynamically reducing the penalty of the ambiguous angular positions according to the aspect ratio, so the network can get a more robust direction estimation result for the square-like objects. As illustrated in Fig. 1(f), the network can output a bimodal angular distribution, which is reasonable for the square-like object. The source code is available at https://github.com/WangJian981002/GF-CSL.

In summary, the main contributions of this work are as follows.

1) We reconstructed the training objective to a Gaussian FL for the classification-based rotated detectors. By adopting a dynamic weighting mechanism, it can promote the network to generate accurate and high confident angle prediction results while avoiding the regression inconsistency problem.
2) We design an aspect ratio-aware factor to enhance the robustness of the model for determining the orientation of square-like objects. It can help the network learn the angle information easier and stabilize the training process.
3) We conduct extensive experiments on aerial image datasets DOTA, HRSC2016, and UCAS-AOD to demonstrate the effectiveness and superior performances of our approaches. Our best model achieves 79.94%, 97.90%, and 96.51% mean average precision (mAP) on DOTA, HRSC2016, and UCAS-AOD, respectively.

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This article is organized as follows. We first briefly recall some related works in Section II. Then, our proposed methodology is introduced in Section III. Section IV presents a comparative study and comparisons with state-of-the-art methods. The conclusions and future works are discussed in Section V.

II. Related Work

A. Horizontal Object Detection Method

Due to the success of CNNs [25]–[29], horizontal object detection in natural images has made tremendous progress during the last decade. In general, CNN-based object detectors can be divided into two-stage methods [8]–[10], [30]–[33] and one-stage methods [11]–[14], [34]–[36].

1) Two-Stage Methods: Regions with CNN features (RCNN) [8] pioneers the region-based two-stage methods, where the first stage is used to generate a sparse set of candidate proposals through selective search algorithms [37] and using CNNs to extract features on the proposals in the second stage. Fast-RCNN [9] improves RCNN by extracting region-of-interest (RoI) from the feature maps, which can save computation from shared backbone networks. A region proposal network (RPN) is first introduced in Faster-RCNN [10] for proposal generation; thus, the entire network can be trained in an end-to-end manner. Mask-RCNN [31] extends Faster-RCNN by adding an extra branch in parallel with the bounding box regression branch to predict an object mask. To detect objects with different scales, a feature pyramid network (FPN) [30] was proposed to build pyramidal hierarchy in deep convolutional networks with marginal extra cost, which can consistently improve the detection performance. Different from densely placing predefined anchors on the image to obtain region proposals, Sparse-RCNN [33] utilizes a fixed sparse set of learnable proposals to replace the RPN, which eliminates the anchor-related designs and many-to-one label assignment.

2) One-Stage Methods: One-stage detectors remove the RoI extraction process and perform bounding box regression and classification simultaneously. Single shot multibox detector (SSD) [11] and you only look once (YOLO) [12] have popularized the one-stage approaches due to their high computational efficiency, where the former densely places anchor boxes over input image, and the latter divides the input into grids of different sizes. To solve the class imbalance problem caused by sparse foregrounds, focal loss [13] was designed to dynamically adjust the weights of each anchor box. The new one-stage detector called RetinaNet can even outperform the two-stage paradigms. Fully convolutional one-stage object detection (FCOS) [14] further improves RetinaNet by removing the predefined anchor boxes and directly performs regression and classification on the reference points. CornerNet [38] is another anchor free approach that directly detects an object using pairwise corners. Inspired by CornerNet, CenterNet [39] proposes to regress the width and the height of the bounding box from the object center, which can avoid the non-maximum suppression process and reduce the inference speed of the detector. Recently, query-based detectors [35], [36] have gradually shown their advantages due to their end-to-end characteristic. DEtection TRansformer (DETR) [35] uses the encoder–decoder structure of the transformer to convert the object detection into a set prediction problem. Deformable DETR [36] designs a deformable attention module to focus on a small set of key sampling locations, thus solving the problem of slow convergence of DETR.

B. Rotated Object Detection Method

Recently, the establishment of a large number of aerial image datasets [1]–[3] has promoted the development of this field. Most rotated detectors are developed by adaption of horizontal detection methods using rotated bounding boxes [16], [40] or quadrangles [17], [41], [42] to represent arbitrary-oriented objects. Instead of using horizontal anchors, [18], [43] and [44] adopt rotated anchors to get more elaborate region proposals. To solve the mismatch issue under densely packed objects, RoI transformer [16] proposes to apply spatial transformations on RoIs and learn the transformation parameters under the supervision of OBBs. Oriented RCNN [45] designs an oriented RPN to directly generate high-quality-oriented proposals in a nearly cost-free manner. Single-shot alignment network (S^2A-Net) [46] proposes a feature alignment module to obtain high-quality anchors and alignment convolution to adaptively align the deep features. Based on Faster-RCNN, context-aware detection network (CAD-Net) [20] was proposed to learn global and local contexts in remote sensing images, which can focus on more informative regions and features through a spatial-and-scale-aware attention module. Scrdet++ [21] designs a novel instance-level denoising (InLD) module over feature map to detect small, cluttered, and rotated objects. Rotation-equivariant detector (ReDet) [47] built a rotation-equivariant detector for high-quality aerial object detection, which encodes both rotation equivariance and rotation invariance.

C. Solution to Boundary Discontinuity

In the rotated object detection process, directly predicting the angle of the target may face the problem of boundary discontinuity, so a lot of researchers are exploring how to mitigate this issue. We summarize them into two categories: start with the representation of the rotated object [48]–[51] and start with the loss function [17], [21]–[24], [52]–[54].

1) Different Representations for Rotated Objects: As the five-parameter representations (c_x, c_y, h, w, θ) may suffer from the periodicity of angular and exchangeability of edges [22] while the eight-parameter representations $(x_1, y_1, x_2, y_2, x_3, y_3, x_4, y_4)$ will face the problem of vertex order, so, researchers have designed lots of instructive representations for rotated objects to circumvent the abovementioned problems. One typical representation method is based on segmentation [48], [49]. The key idea is to seek the object region at pixel level and then use the postprocessing methods to obtain the OBBs, which can largely eliminate the ambiguities on the target definitions and the background pixels. Liu *et al.* [50] discretized the OBB into a series of key edges without the sequential relationship, which can not only avoid the learning confusion issue but also improve the scene text detection performance. Yi *et al.* [51] performed the detection process using four box boundary-aware vectors. The proposed methods, called BBAVectors, capture the vectors,

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which are distributed in the four quadrants of a Cartesian coordinate system in an anchor-free way. Gaussian Wasserstein distance (GWD) [53] converts the rotated bounding box to a 2-D Gaussian distribution and uses GWD as a loss function, which can approximate the indifferentiable rotated intersection over union (IoU) calculation. KLD [54] further boosts the GWD using Kullback–Leibler divergence.

2) Loss Functions for Boundary Discontinuity: Another type of approach is to eliminate the sudden increase of loss in boundary cases at the loss function level. Textboxes++ [17] uses such a way to determine the order of the vertices under eight-parameter representation: the four vertices are organized in a clockwise order, such that the sum of Euclidean distances with reference points is minimum. Rotation sensitivity detector (RSDet) [52] learns a modulated loss to deal with the rotation sensitivity error both for five-parameter and eight-parameter methods. Scrdet++ [21] designs a novel IoU constant factor based on the smooth $L1$ loss to address the boundary problem. The abovementioned methods only alleviate the boundary discontinuity problem but do not effectively solve it. Yang and Yan [22] innovatively transformed the angle regression problem into a classification task and used the CSL to solve the boundary discontinuity, which directly contributes to the development of classification-based rotated object detection algorithms. Densely coded labels (DCL) [23] extends CSL by adopting a loss reweighting strategy to improve the detection accuracy for square-like objects.

III. Methods

A. Analysis of FL for Binary Soft Label

We first briefly introduce the CSL for angular classification and then analyze the limitations of FL for the binary soft label.

We assume that the angle takes a range of values T ($T = 180°$ under five-parameter long side definition), and ΔT degree ($\Delta T = 1°$ by default) per interval refers to a category. Direct use of one-hot label cannot provide certain fault tolerance for adjacent and cross-boundary angle categories, so the label is designed as a periodic smooth label. Fig. 1(a) and (d) shows the examples of CSL, which adopt Gaussian function for label smoothing. Because CSL has the characteristics of periodicity and smoothness, it can naturally solve the problem of discontinuous angle boundaries while having a certain error tolerance for the prediction of adjacent angle categories.

Then, we introduce the loss function for the binary soft label. FL [13] was originally designed to solve the class imbalance problem between foreground and background samples, which can be formulized as follows:

$$\mathrm{FL}(x) = -\alpha_t(1-x_t)^{\gamma}\log(x_t), \quad x_t = \begin{cases} x, & \text{if } y = 1 \\ 1-x, & \text{if } y = 0 \end{cases} \tag{1}$$

where $y \in \{0, 1\}$ denotes the one-hot label at each category, and $x \in [0, 1]$ is the network's estimated probability for each class. α_t is a weighting factor, which is equal to $\alpha \in [0, 1]$ for class 1 and $1-\alpha$ for class 0. The current form of FL can only support $\{0, 1\}$ discrete labels, and the direct expansion to the continuous form is as follows:

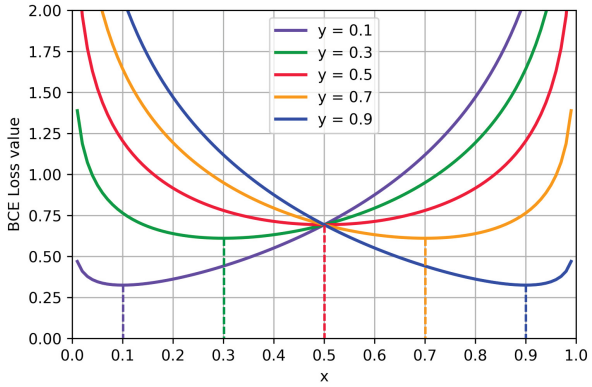


Fig. 2. BCE loss value under several label y and corresponding global minimum values are marked with dashed lines.

$$\mathrm{FL}(x) = \alpha_t \cdot (1-x_t)^{\gamma} \cdot \mathrm{BCE}(x, y) \tag{2}$$

$$L_{\mathrm{FL}} = \frac{1}{|X|}\sum_{x \in X} \mathrm{FL}(x) \tag{3}$$

where $X \in [0, 1]^{(T/\Delta T)}$ is the model's estimated angular distribution, and the detailed expressions of each item are modified as follows:

$$\alpha_t = y\alpha + (1-y)(1-\alpha) \tag{4}$$

$$x_t = yx + (1-y)(1-x) \tag{5}$$

$$\mathrm{BCE}(x, y) = -((1-y)\log(1-x) + y\log(x)). \tag{6}$$

Now, the continuous FL can be directly used for soft label $y \in [0, 1]$. Note that the BCE item in continuous FL has a global minimum when the predicted class score x is equal to the label y (as illustrated in Fig. 2). So, the model is encouraged to estimate an intermediate value associated with soft label y for each angle class (detailed proof can be found in Appendix A). It is difficult to accurately predict this artificial value at each location. When loss values are summed over all the negative angle categories, especially for some confusing positions such as categories near the target angle interval, these relatively insignificant loss contributions can overwhelm the target position, thus preventing the network from predicting precise angle information. Intuitively, we show some loss comparisons for several simulated angular distributions in Fig. 3, which adopts continuous FL as comparison criteria. Comparing X_2 with X_3, both have accurate angular prediction results (the position of angular peak). However, X_2 has a higher prediction confidence but the loss value is larger than X_3. Comparing X_1 with X_2, the peak position of X_1 is not as accurate as X_2, but has a smaller loss value. So, the continuous version of FL is not optimal for the angular classification.

B. Gaussian FL for Angular Classification

To learn accurate and high confident angle estimations while avoiding the boundary discontinuity problem, we reconstruct the training objective to a Gaussian FL inspired by [38]. The complete loss function, which we called Gaussian focal-CSL

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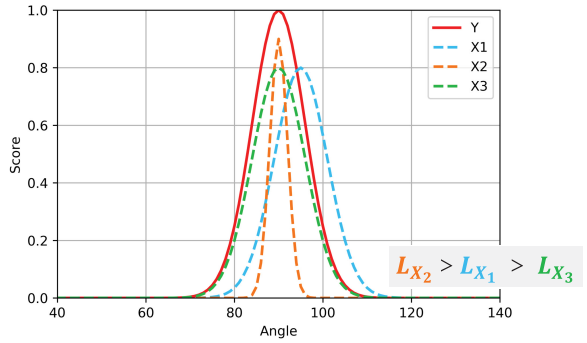


Fig. 3. Loss comparisons of several simulated angle distributions $\{X_1, X_2, X_3\}$. The loss function adopts continuous FL. Red solid line indicates the soft label Y, and the dashed lines indicate the predicted distribution.

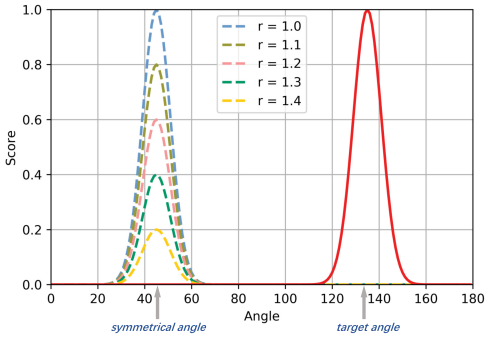


Fig. 4. Modulated smooth label $\tilde{Y}$ under different aspect ratios. The red solid line indicates CSL Y at target position, and the dashed lines indicate $\tilde{Y_{90}}$ at symmetrical position. The combination (element-wised maximum) of Y and $\tilde{Y_{90}}$ is the full smooth label $\tilde{Y}$.

(GF-CSL), can be formulized as follows:

$$\begin{aligned}&\text{GF_CSL}(X, Y)\\&= -\frac{1}{|X|}\sum_{x\in X}\begin{cases}(1-x_t)^{\gamma}\log(x_t), & \text{if } y = 1\\(1-y)^{\beta}(1-x_t)^{\gamma}\log(x_t), & \text{otherwise}\end{cases}\end{aligned} \tag{7}$$

where $|X| = (T/\Delta T)$ is the number of angle category. We now explain how it was designed.

The loss function needs to satisfy the following three-part objectives: first, the ultimate goal of optimization is to obtain a precise angle, that is, an accurate peak position and higher peak response; second, it has to be circular to adapt to discontinuous angle boundaries while having certain error tolerance between adjacent angle categories; third, the network is more focused on the foreground category. The abovementioned objectives can be achieved by the following components, respectively.

1) The cross-entropy part $-\log(x_t), x_t = x$ only if $y = 1$; otherwise, $x_t = 1 - x$. Unlike the BCE part that encourages a continuous value close to y, we only stimulate a higher response at the target category and suppress the rest.
2) The Gaussian reweight part $(1 - y)^{\beta}$ applied to the negative locations ($y \neq 1$), while $y \in Y$ is the CSL at each category. In the training process, the loss weight is adaptively reduced as the distance from the target category becomes smaller, so the network has a certain error tolerance for the prediction which is close to the positive category. At the same time, the circular of Y avoids the interference of discontinuous boundaries.
3) The modulating factor part $(1 - x_t)^{\gamma}$, which is used to balance the number of angle categories for positive and negative samples. It can reshape the loss function to down-weigh negative categories and, thus, focus training on positive angle location.

To deal with adjacent and cross-boundary angle categories, our GF-CSL uses a dynamic weighting approach on all negative angular categories, while continuous FL employs a predictive approach that predicts the relevant soft label value at each position. GF-CSL can promote the network to learn a more polarized and accurate angle distribution while avoiding the sudden increase in loss value caused by the discontinuous boundary.

C. Aspect Ratio-Aware Factor for Square-Like Object

The widely used five parameters $[c_x, c_y, h, w, \theta]$-oriented representation may fail for the square-like objects. We analyze that the main reason for this phenomenon is the larger loss value generated at a position 90° different from the target angle category. We refer to this position as the symmetrical angle for simplicity. For an ideal square object, both the target angle and the symmetrical angle can accurately describe the direction of that. So, when the aspect ratio (we refer to the long side divided by the short side specifically in this article) of an object is close to 1, the responses at these two positions should be stimulated simultaneously. The confusing effects caused by symmetrical locations should gradually weaken as the aspect ratio of the object increases.

Based on the analysis, we proposed an aspect ratio-aware factor applied to GF-CSL to smoothly eliminate ambiguity at symmetrical angles for square-like objects. An intuitive approach is to dynamically reduce the weight of the classification loss at the symmetrical position according to the aspect ratio of the target. As the CSL Y in $(1 - y)^{\beta}, y \in Y$ part is used to adjust loss weights around target categories, so it can be naturally extended to the symmetrical angle position. In particular, let Y_{90} denote the label after Y is translated by 90°; that is, Y_{90} is the CSL at the symmetrical angle. Then, an aspect ratio factor is acted on Y_{90}

$$\tilde{Y_{90}} = g(r)Y_{90} \tag{8}$$

where $r = (h/w)$ is the aspect ratio, and $g(\cdot)$ is a monotonical decreasing function that satisfies $g(1) = 1$ and $g(r_{\text{thr}}) = 0$. r_{thr} is the maximum aspect ratio threshold determining whether an object is needed to adjust the loss weight at symmetrical positions. The final modulated smooth label $\tilde{Y}$ used in $(1 - y)^{\beta}, y \in \tilde{Y}$ can be formulized as

$$\tilde{Y} = \text{Element_wised_maximum}\left(Y, \tilde{Y_{90}}\right). \tag{9}$$

Fig. 4 shows several $\tilde{Y}$ values under different aspect ratios when $r_{\text{thr}} = 1.5$. When the aspect ratio $r = 1$, the reweight part $(1 - y)^{\beta}$ decays to zero at the symmetrical angle, which indicates that the model is allowed to produce a higher response at symmetrical positions. When r increases to r_{thr}, $\tilde{Y}$ degenerates into Y. It is worth noting that when the aspect

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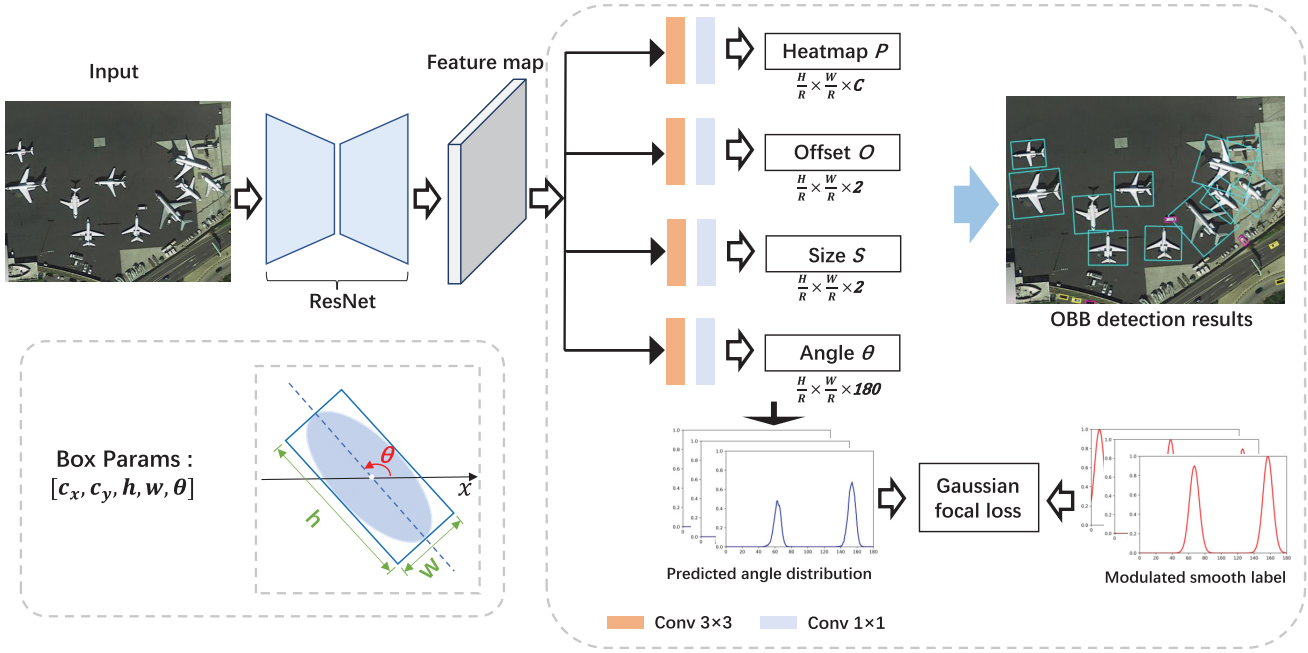


Fig. 5. Overall architecture of our method to detect arbitrary-oriented objects in aerial images. Our proposed Gaussian FL is used for angle classification.

ratio of the object is relatively large, its angle information is relatively clear, so the r_{thr} cannot be set too large to ensure accurate angle prediction.

D. Overall Architecture

Compared with the regression-based rotated methods, the classification-based detectors require relatively heavy prediction layers for angle estimation, especially the anchor-based design will increase the complexity of the model more seriously, because it is necessary to perform angle classification calculations for each anchor. To improve the overall efficiency of the detection model, we choose the keypoint-based anchor free method [39] to further build our rotated detector. The overall architecture is illustrated in Fig. 5. In particular, the heatmap branch is utilized to detect the center points of arbitrarily oriented objects in aerial images. Several parallel box parameter branches are used to determine the OBBs for each object under five parameters definition $[c_x, c_y, h, w, \theta]$.

1) Backbone: We use ResNet as backbone and build a pyramidal feature hierarchy on the network's output features $\{C_3, C_4, C_5\}$ that follow [39] and [55]. The intermediate features $\{C_3, C_4, C_5\}$ have receptive fields of $\{8, 16, 32\}$ pixels and corresponding channels $\{64, 128, 256\}$, $\{128, 192, 256\}$, and $\{256, 256, 256\}$ for ResNet50, ResNet101, and ResNet152, respectively.

2) Center Heatmap: The heatmap is used to localize the center points of oriented objects in the input image. The network estimates a center heatmap $P \in [0, 1]^{(H/R)\times(W/R)\times C}$ for each input, where H and W are the height and the width of the input image. R is the stride between heatmap P and input, which is set to 4 following [39]. The predicted $P_{xyc} = 1$ denotes a detected center point of the rotated object.

Following [38] and [39], we splat all ground-truth center points onto a heatmap $\hat{P} \in [0, 1]^{(H/R)\times(W/R)\times C}$ and pass through a Gaussian kernel. The training objective is

$$L_c = -\frac{1}{N}\sum_{x,y,c}\begin{cases}(1-P_{xyc})^{\gamma}\log(P_{xyc}), & \text{if } \hat{P}_{xyc}=1\\ (1-\hat{P}_{xyc})^{\beta}{P_{xyc}}^{\gamma}, & \text{otherwise}\\ \log(1-P_{xyc})\end{cases} \tag{10}$$

where N is the number of objects in the image. We use $\gamma = 2$ and $\beta = 4$ following [39], which is the same as (7). To eliminate the discretization error caused by the output stride, we predict an additional center offset $O \in [0, 1]^{(H/R)\times(W/R)\times 2}$, where the standard Smooth $L1$ [9] is applied during training.

3) Box Parameters: The long side h and short side w of each rotated object are predicted from an object size branch $S \in \mathbb{R}^{(H/R)\times(W/R)\times 2}$. Different from [39] directly predicts the raw pixel of $\{h, w\}$, we predict an encoded form $\{\ln((h/R)), \ln((w/R))\}$ to reduce the impact of object scale changes. The Smooth $L1$ is also used for size estimation.

For the object orientation, the network outputs a $\theta \in [0, 1]^{(H/R)\times(W/R)\times 180}$ for angle classification, where our proposed Gaussian FL is applied for training the orientation branch.

IV. Experiments

A. Datasets and Implementation Details

1) DOTA: DOTA [1] is a large-scale dataset for rotated object detection in aerial images, where the images are collected from different sensors and platforms. It provides 2806 aerial images with totally 188 282 instances of 15 object categories: plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer-ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). There are 1411, 458, and 937 images for training, validation, and

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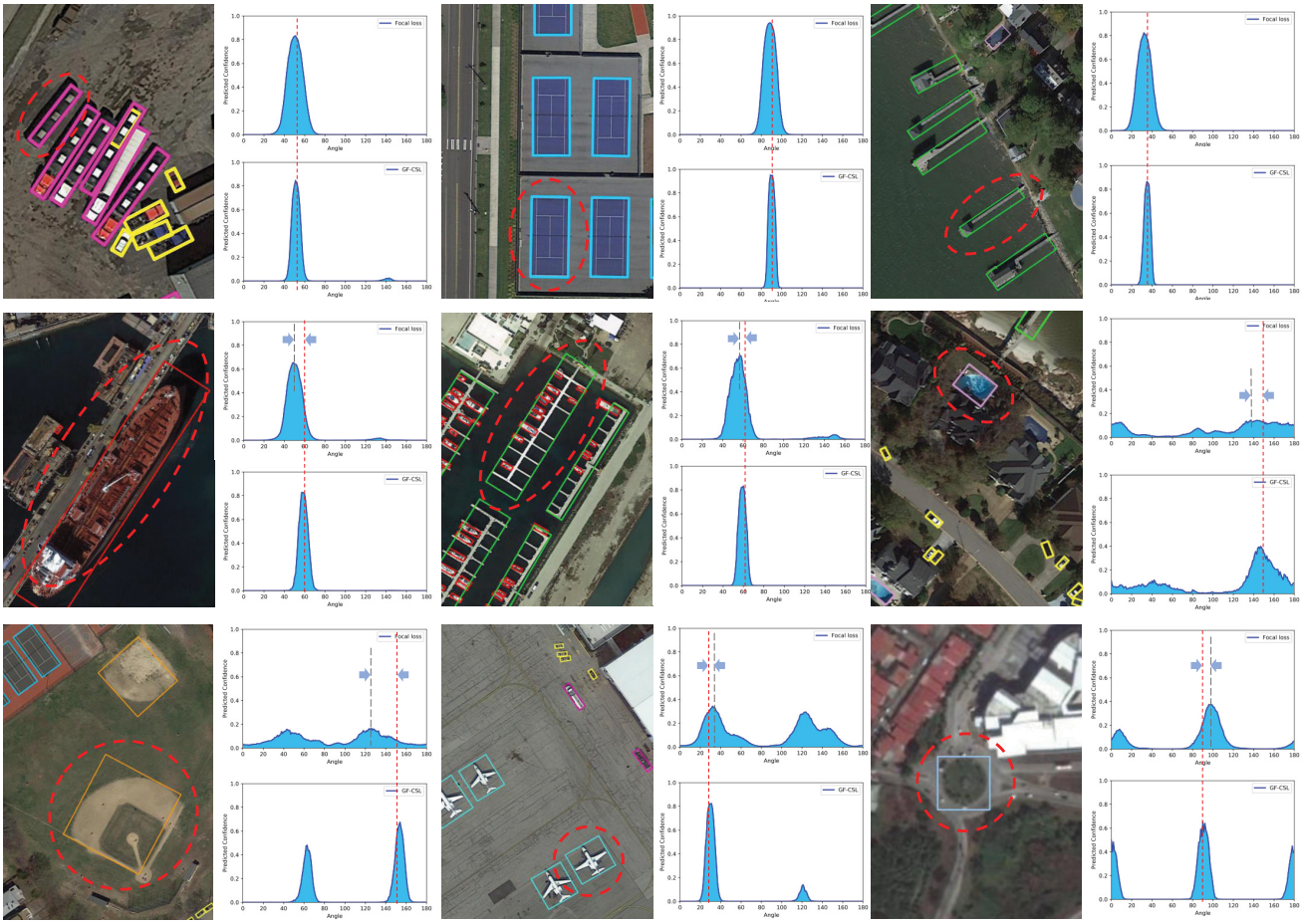


Fig. 6. Visualization of estimated angle distributions from the DOTA dataset. The objects to be shown in each image have been marked with circles. In each subimage, the top result is based on FL, while the bottom one is based on GF-CSL. The ground-truth angle values of each object have been marked with a red dashed line. The detection results for each image are based on GF-CSL.

testing, respectively. The size of each image ranges from 800 × 800 to 4000 × 4000. To fit the limited GPU memory, we crop training images into the patches of size 1024 × 1024 pixels with an overlap of 256 pixels, which leads to 14 560 labeled images in the training stage. During testing, we crop the test images into 4000 × 4000 patches with an overlap of 1024 pixels to reduce detection errors at the cutting edge.

2) HRSC2016: HRSC2016 [2] is dedicated to SH detection in aerial images, which contains 1061 images from two scenarios, including SHs at sea and SHs inshore at six famous HAs. There are 436, 181, and 444 images for training, validation, and testing, respectively, with the image size ranging from 300 × 300 to 1500 × 900. We resize the long side of each image to a fixed size (e.g., 640 px, detailed at Section IV-D) and keep the original aspect ratio both for training and testing.

3) UCAS-AOD: UCAS-AOD [3] is a publicly accessible dataset for rotated object detection including two categories: car and PL. It contains 1510 aerial images with totally 14 595 objects. The resolution of all images is 1280 × 659 pixels. Following [22], [23], and [49], we randomly split the original dataset into 1110 and 400 for training and testing, respectively. We adopt the same data processing strategy as HRSC2016.

4) Evaluation Metric: The widely used mAP is adopted to evaluate the performance of detectors in all our experiments, which is generally defined as follows:

$$\text{mAP} = \frac{1}{C}\sum_{c=1}^{C}\int P_c(R_c)\,d\,R_c \tag{11}$$

where P_c and R_c denote the precision and the recall for each category c, respectively. For the DOTA dataset, the test results are accessed from the official evaluation server. For HRSC2016 and UCAS-AOD, we report the VOC12 and VOC07 AP metrics with a IoU threshold of 0.5.

5) Implementation Details: All experiments are conducted in PyTorch on GeForce RTX 3090 GPUs. We use ImageNet-pretrained ResNet [28] as our backbone. The whole networks are optimized by Adam for 140 epochs with the learning rate dropped by 10× at 100 and 130 epochs. As the DOTA dataset takes a relatively large image resolution as an input, the batch size is set for 12 with an initial learning rate 1.25×10^{-4}. For the HRSC2016 and UCAS-AOD datasets, the batch size is uniformly set to 32 with initial learning rates set as 2×10^{-4} and 1×10^{-4}, respectively. The monotonical decreasing function $g(\cdot)$ used in (8) takes a linear function, and r_{thr} is set to 1.4 by default. We perform data augmentations, including random horizontal and vertical flipping (with probability 0.5), random graying (with probability 0.1), and random rotation (with maximum rotation angle 10°). By default, ResNet50 is

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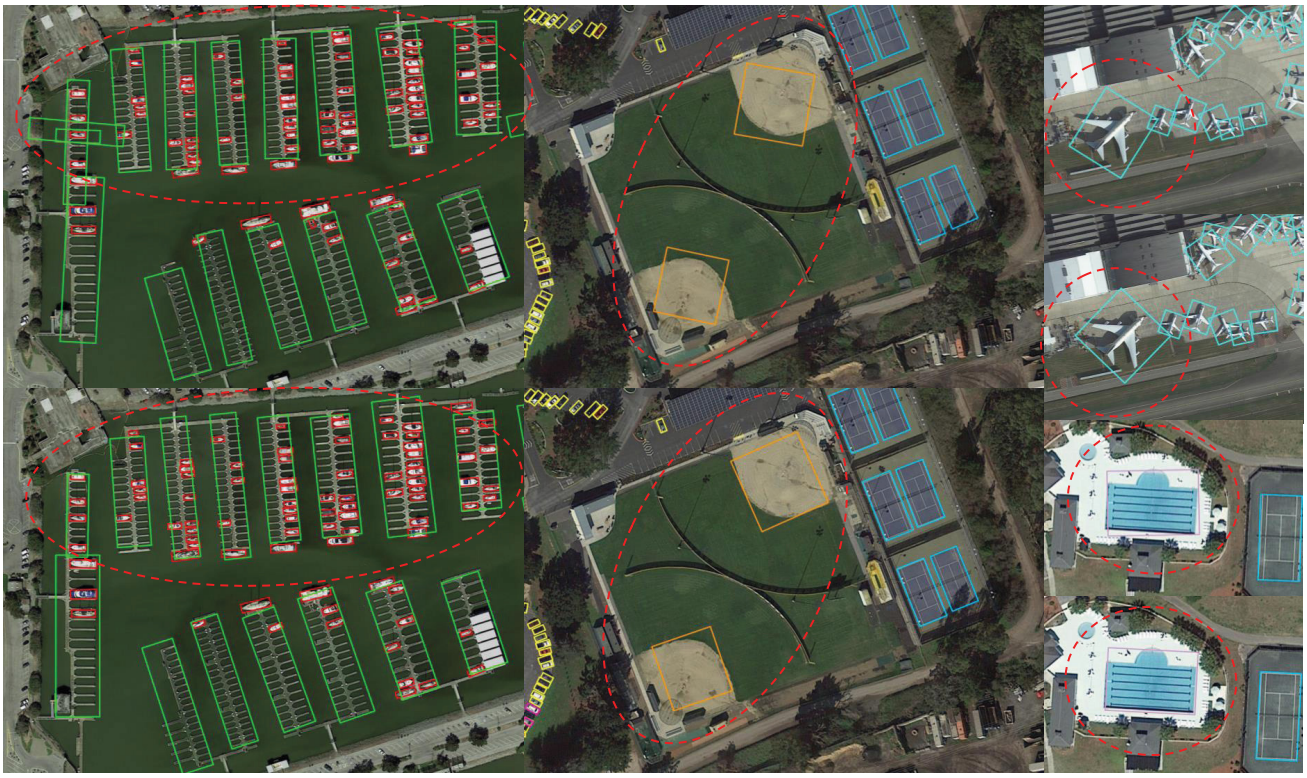


Fig. 7. More visual comparisons on the DOTA dataset. For each pair of detection results, the top one is based on FL, and the bottom one is based on Gaussian FL (GF-CSL).

TABLE I

ABLATION STUDIES OF OUR PROPOSED METHODS WITH DIFFERENT WAYS FOR ANGLE ESTIMATION. THE RESULTS ARE EVALUATED ON THE TEST SET OF DOTA FOR ORIENTED OBJECT DETECTION TASK

Method	Backbone	mAP (%)	Improvement
Reg	ResNet50	73.05	+ 0.00
CSL	ResNet50	73.68	+ 0.63
GF	ResNet50	74.02	+ 0.97
GF+AS	ResNet50	74.68	+ 1.63

used for ablation studies and deeper networks for state-of-the-art comparisons.

B. Ablation Study

Due to the rich scenarios and diverse categories of the DOTA dataset, we use DOTA as our main dataset for ablations.

1) Effectiveness of Gaussian FL With Aspect Ratio-Aware Factor: To explore the effectiveness of our proposed methods, we conduct several baseline experiments, including regression-based angle prediction (Reg), CSL-based angle classification with continuous FL (CSL), Gaussian FL for angle classification (GF), and Gaussian FL with aspect ratio-aware factor (GF + AS). The abovementioned methods share the same network structure except for the different orientation branches.

The results are shown in Table I, and using a regression way to directly predict the angle of the rotated object achieves 73.05% mAP on the DOTA test set. As the angular classification-based CSL eliminates the discontinuous boundary issues caused by the angular periodicity, it improves the baseline method by 0.63% mAP and achieves 73.68%. By reconstructing the training objective to a Gaussian FL, the model will be more dedicated to predicting precise object angles. Thus, our GF achieves 74.02% mAP on the DOTA test set, which surpasses the CSL competitor by 0.34% mAP. Since the aspect ratio-aware factor is friendly to square-like objects and can be flexibly applied to the Gaussian FL, the combination of GF + AS further boost the GF by 0.66% mAP and achieves 74.68% on DOTA. Table II directly demonstrates that our aspect ratio-aware factor can effectively handle square-like objects on several categories with small aspect ratios. To further evaluate the angle prediction accuracy of our methods, we calculate the mean average orientation error (mAOE°) across all categories on the DOTA validation set following [67]. Table III shows that our Gaussian FL can significantly reduce the angle prediction error. In the rest of our paper, we call our classification-based rotated detector equipped with Gaussian FL and aspect ratio-aware factor as GF-CSL for simplicity.

TABLE II

EXPERIMENTAL RESULTS ON SEVERAL SMALL ASPECT RATIO CATEGORIES IN THE DOTA DATASET. THE ANGLE CLASSIFICATION-BASED DETECTORS ARE PRESENTED, AND 5 mAP MEANS THE PERFORMANCE OF THE FIVE CATEGORIES LISTED

Method	PL	GTF	ST	SBF	RA	5-mAP	mAP
CSL	89.60	64.57	84.28	54.53	60.61	70.72	73.68
GF	89.82	65.95	84.11	52.70	59.64	70.44	74.02
GF+AS	89.67	70.18	84.43	57.69	63.49	73.09	74.68

2) Influence of Aspect Ratio Threshold: In our GF-CSL, the aspect ratio threshold r_{thr} is used to decide whether an object is needed to adjust the loss weight at symmetrical positions. If the aspect ratio of an object is less than r_{thr},

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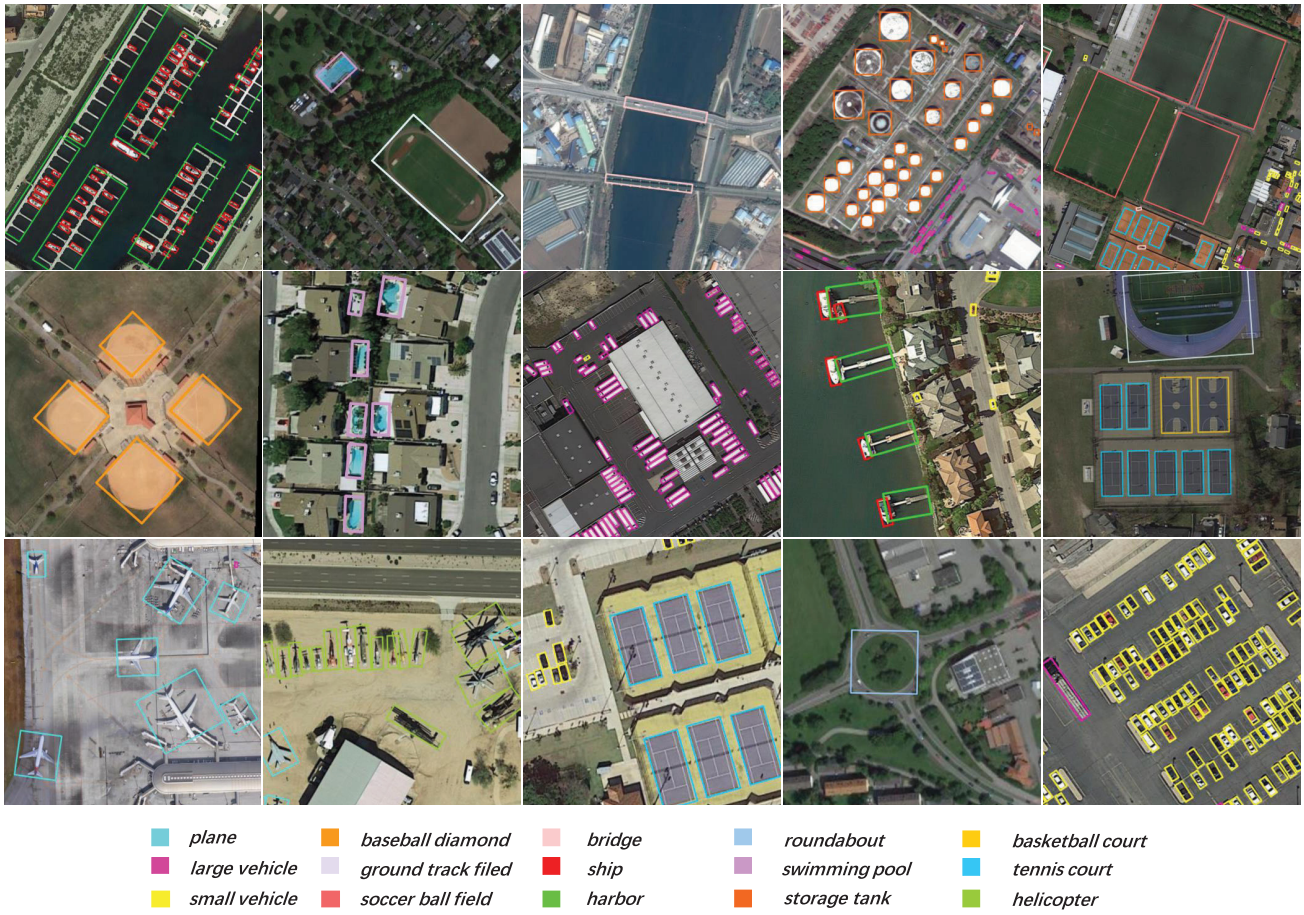


Fig. 8. Visualization results of our GF-CSL on the DOTA dataset.

TABLE III

MEAN AVERAGE ORIENTATION ERRORS ON DOTA VALIDATION SET

Method	Reg	CSL	GF	GF+AS
mAOE°	8.17	8.12	5.92	5.68

TABLE IV

mAPs OF DIFFERENT VALUES OF ASPECT RATIO THRESHOLD r_{THR} ($g(\cdot)$ IS A LINEAR DECAY FUNCTION)

r_{thr}	PL	GTF	ST	SBF	RA	5-mAP	mAP
1.0	89.82	65.95	84.11	52.70	59.64	70.44	74.02
1.1	89.88	73.52	84.78	49.48	60.29	71.59	74.17
1.2	89.53	72.33	84.90	51.73	61.07	71.91	74.22
1.3	89.55	69.93	83.50	52.78	60.97	71.35	74.45
1.4	89.67	70.18	84.43	57.69	63.49	**73.09**	**74.68**
1.5	89.99	69.78	84.55	50.43	63.56	71.66	74.59

TABLE V

mAPs OF DIFFERENT TYPES OF ATTENUATION FUNCTION $g(\cdot)$ ($r_{\text{THR}} = 1.4$)

Type	PL	GTF	ST	SBF	RA	5-mAP	mAP
Linear	89.67	70.18	84.43	57.69	63.49	**73.09**	**74.68**
Cosine	89.87	71.61	84.08	52.34	66.21	72.82	74.49
Step	89.81	72.41	84.25	52.41	64.53	72.68	74.42

the loss penalty at its symmetrical position will be reduced accordingly. A larger r_{thr} can take care of objects with a wider range of aspect ratios; however, r_{thr} cannot be set too large to prevent the wrong angle prediction at the symmetrical positions. As shown in Table IV, the best performance can be obtained when r_{thr} is set to 1.4. When r_{thr} is too large, it will affect the accurate prediction of the angle. When r_{thr} is too small, the effect at the symmetrical positions is not obvious. Especially when $r_{\text{thr}} = 1.0$, GF + AS will degenerate to GF.

3) Influence of Attenuation Function: The attenuation function [in (8)] gradually weakens the influence at the symmetrical positions according to the object aspect ratio (from 1 to r_{thr}). Three kinds of decay functions $g(\cdot)$, including linear function, cosine function, and step function, are introduced for investigation. Table V shows the performance comparison of three attenuation functions on the DOTA dataset. It can be seen that the linear decay performs best, while the step function performs worst because of its discontinuity point.

4) Visualization of Estimated Angle Distribution: We visualize some of the predicted angular distributions shown in Fig. 6, which includes CSL with continuous FL and our GF-CSL. In the first row of Fig. 6, we selected some simple cases with clear angle information, so that both GF-CSL and FL can achieve accurate prediction results. However, our method can obtain higher peak responses. In the second row of Fig. 6, we selected some difficult samples, which may be attributed to large scale differences, complex backgrounds, or insignificant orientation information of the target. From the results, we can observe that GF-CSL can get more precise classification results (the position of angular peak). In the third row of Fig. 6, we selected some square-like objects, and

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TABLE VI

STATE-OF-THE-ART COMPARISONS ON THE TEST SET OF DOTA FOR THE OBB TASK. DETECTION ACCURACIES FOR EACH CATEGORY (AP) AND OVERALL PERFORMANCE (MAP) ARE PROVIDED. MS INDICATES THAT MULTISCALE TRAINING AND TESTING ARE USED

Methods	Backbone	MS	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
Two-stage methods																		
DPSRP [16]	ResNet101	√	81.18	77.42	35.48	70.41	56.74	50.42	53.56	89.97	79.68	76.48	61.99	59.94	53.34	64.04	47.76	63.89
ICN [15]	ResNet101	√	81.40	74.30	47.70	70.30	64.90	67.80	70.00	90.80	79.10	78.20	53.60	62.90	67.00	64.20	50.20	68.20
ROI-Trans [16]	ResNet101	√	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.34	62.83	58.93	47.67	69.56
CAD-NET [20]	ResNet101		87.80	82.40	49.40	73.50	71.10	63.50	76.70	90.90	79.20	73.30	48.40	60.90	62.00	67.00	62.20	69.90
FFA-3+ [56]	ResNet101	√	89.60	76.40	48.20	58.90	67.20	76.50	81.40	90.10	83.30	83.40	55.70	60.20	73.20	66.70	64.90	71.80
Cascade-FF [57]	ResNet152		89.90	80.40	51.70	77.40	68.20	75.20	75.60	90.80	78.80	84.40	62.30	64.60	57.70	69.40	50.10	71.80
Scrdet [58]	ResNet101	√	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	87.94	86.86	65.02	66.68	66.25	68.24	65.21	72.61
FADet [59]	ResNet101	√	90.21	79.58	45.49	76.41	73.18	68.27	79.56	90.83	83.40	84.68	53.40	65.42	74.17	69.69	64.86	73.28
Gliding Vertex [60]	ResNet101		89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32	75.02
Mask OBB [48]	ResNeXt101	√	89.56	85.95	54.21	72.90	76.52	74.16	85.63	89.85	83.81	86.48	54.89	69.64	73.94	69.06	63.32	75.33
FFA-3++ [56]	ResNet101	√	90.10	82.70	54.20	75.20	71.00	79.90	83.50	90.70	83.90	84.60	61.20	68.00	70.70	76.00	63.70	75.70
CenterMap [49]	ResNet101	√	89.93	84.41	54.60	70.25	77.66	78.32	87.19	90.66	84.89	85.27	56.46	69.23	74.13	71.56	66.06	76.03
CSL-FPN [22]	ResNet152	√	90.25	85.53	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	69.60	68.04	73.83	71.10	68.93	76.17
Scrdet++ [21]	ResNet101	√	90.05	84.39	55.44	73.99	77.54	71.11	86.05	90.67	87.32	87.08	69.62	68.90	73.74	71.29	65.08	76.81
FR-Est-FPN [61]	ResNet101	√	89.77	85.16	55.32	76.47	80.08	83.26	87.53	90.61	87.36	86.89	65.09	68.48	71.63	79.60	66.10	77.62
DEA-ROI Trans [62]	ResNet101	√	89.18	83.46	55.14	71.69	79.59	83.08	88.10	90.88	87.09	86.73	63.99	65.14	75.81	78.01	68.69	77.77
ReDet [47]	ReR50	√	88.81	82.48	60.83	80.82	78.34	86.06	88.31	90.87	88.77	87.03	68.65	66.90	79.26	79.71	74.67	80.10
Oriented R-CNN [45]	ResNet50	√	89.84	85.43	61.09	79.82	79.71	85.35	88.82	90.88	86.68	87.73	72.21	70.80	82.42	78.18	74.11	**80.87**
Single-stage methods																		
TOSO [63]	ResNet101	√	80.17	65.59	39.82	39.95	49.71	65.01	53.58	81.45	44.66	78.51	48.85	56.73	64.40	64.24	36.75	57.92
PIou [64]	DLA-34		80.90	69.70	24.10	60.20	38.30	64.40	64.80	90.90	77.20	70.40	46.50	37.10	57.10	61.90	64.00	60.50
BBAVectors [51]	ResNet101	√	88.35	79.96	50.69	62.18	78.43	78.98	87.94	90.85	83.58	84.35	54.13	60.24	65.22	64.28	55.70	72.32
RSDet [52]	ResNet152		90.10	82.00	53.80	68.50	70.20	78.70	73.60	91.20	87.10	84.70	64.30	68.20	66.10	69.30	63.70	74.10
CenterRot [55]	ResNet152		89.69	81.42	51.16	68.82	78.77	81.45	87.23	90.82	80.31	84.27	56.13	64.24	75.80	74.68	56.51	74.75
DA-Net [65]	ResNet101	√	89.70	82.41	53.28	69.55	78.24	79.54	89.04	90.68	84.76	86.33	65.03	65.70	76.16	73.37	58.86	76.11
R3Det [19]	ResNet152	√	89.80	83.77	48.11	66.77	78.76	83.27	87.84	90.82	85.38	85.51	65.67	62.68	67.53	78.56	72.62	76.47
PolarDet [66]	ResNet101	√	89.65	87.07	48.14	70.97	78.53	80.34	87.45	90.76	85.63	86.87	61.64	70.32	71.92	73.09	67.15	76.64
DCL-R3Det [23]	ResNet152	√	89.26	83.60	53.54	72.76	79.04	82.56	87.31	90.67	86.59	86.98	67.49	66.88	73.29	70.56	69.99	77.37
GWD-R3Det [53]	ResNet152	√	89.28	83.70	59.26	79.85	76.42	83.87	86.53	89.06	85.53	86.50	73.04	67.56	76.92	77.09	71.58	79.08
S^2A-Net [46]	ResNet50	√	88.89	83.60	57.74	81.95	79.94	83.19	89.11	90.78	84.87	87.81	70.30	68.25	78.30	77.01	69.58	79.42
KLD-R3Det [54]	ResNet152	√	89.92	85.13	59.19	81.33	78.82	84.38	87.50	89.80	87.33	87.00	72.57	71.35	77.12	79.34	78.68	**80.63**
GF-CSL (ours)	ResNet50	√	89.77	86.15	49.85	73.01	78.56	81.77	87.95	90.84	87.91	86.23	63.04	66.24	77.14	79.27	65.32	77.54
GF-CSL (ours)	ResNet101	√	89.81	86.64	54.90	76.14	77.96	83.83	87.46	90.85	88.35	86.40	67.25	67.25	77.14	78.25	62.81	78.34
GF-CSL (ours)	ResNet152	√	90.07	84.34	51.08	73.20	79.71	82.73	87.60	90.85	88.00	85.95	65.75	66.10	77.22	79.18	69.98	78.12
GF-CSL* (ours)	ResNet101	√	89.87	85.20	53.24	75.48	79.13	83.59	88.45	90.88	88.28	86.97	66.37	72.64	82.11	82.19	74.63	**79.94**

* The training images are resized into three scales {0.5,1.0,1.5} before cropped into patches following ReDet [47], Oriented R-CNN [45] and S^2A-Net [46].

the results demonstrate the effectiveness of our GF-CSL for square-like objects. Taking the case of the third row and the third column as an example, this angular distribution contains two main peak positions in general: one around 90° and the other around 0°, which is consistent with our analysis for square-like objects. Although each main peak for this example contains some secondary peaks, the highest response is still obtained at 0° and 90°. We also perform some direct visual comparisons shown in Fig. 7.

C. State-of-the-Art Comparison on DOTA

To validate the effectiveness of our approach with other state-of-the-art methods, multiscale training and testing settings are used. We resize the original images at two scales {0.5, 1.0} before dividing the images into patches. Then, we crop resized images into the patches of size 1024 × 1024 pixels with an overlap of 256 pixels and obtain 20 475 labeled images for training. The results are shown in Table VI, and the performances of all methods are taken from a single model without cross-model test augmentation such as model ensemble by default. Our best model gets 79.94% mAP for oriented object detection task on DOTA and achieves best results among all published single-stage methods except for refined single-stage detector with Kullback–Leibler divergence (KLD-R3Det), which adopts model ensemble as augmentation. Moreover, our single-stage approach shows competitive performance against current best two-stage methods, such as ReDet and Oriented RCNN. We visualize some qualitative detection results shown in Fig. 8.

TABLE VII

STATE-OF-THE-ART COMPARISONS ON THE TEST SET OF HRSC2016 FOR ORIENTED OBJECT DETECTION IN AERIAL IMAGES. THE EVALUATION METRICS mAP07 AND mAP12 ARE REPORTED, AND THE RESULTS ARE RANKED BY mAP12

Methods	Backbone	Training size	mAP07	mAP12
ROI-Trans [16]	ResNet101	800*512	86.20	-
Gliding Vertex [60]	ResNet101	800*512	88.20	-
BBAVectors [51]	ResNet101	608*608	88.60	-
CenterMap [49]	ResNet50	-	-	92.80
RetinaNet-R [19]	ResNet101	800*800	89.18	95.21
R3Det [19]	ResNet101	800*800	89.26	96.01
CSL-FPN [22]	ResNet101	-	89.62	96.10
DCL-R3Det [23]	ResNet101	800*800	89.46	96.41
GWD-R3Det [53]	ResNet101	-	89.85	97.37
KLD-R3Det [54]	ResNet101	512*512	89.97	-
Oriented RCNN [45]	ResNet101	800*1333	90.50	97.60
ReDet [47]	ReR50	800*512	90.46	97.63
DEA-ROI Trans [62]	ResNet101	1024*1024	**90.56**	**-**
GF-CSL	ResNet50	512*512	89.80	96.26
GF-CSL	ResNet50	640*640	90.33	97.38
GF-CSL	ResNet101	800*800	90.53	**97.90**

D. Results on HRSC2016 and UCAS-AOD

1) HRSC2016: The HRSC2016 dataset contains lots of SHs with large aspect ratios, so there are still challenges in obtaining accurate detection results. The comparison with other state-of-the-art methods is shown in Table VII, while the commonly used multiscale testing technique is applied during inference. We can see that the strong single-stage detector R3Det achieves the performance of 89.26% on mAP07 and 96.01% on mAP12, which is already an inspiring result for the HRSC2016 dataset. However, our GF-CSL outperforms R3Det

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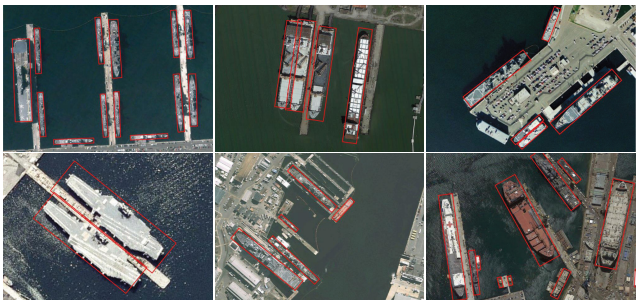


Fig. 9. Visualization results of our GF-CSL on the HRSC2016 dataset.



Fig. 10. Visualization results of our GF-CSL on UCAS-AOD dataset.

TABLE VIII

STATE-OF-THE-ART COMPARISONS ON UCAS-AOD. AP 07/12 FOR EACH CATEGORY AND THE OVERALL mAP 07/12 ARE REPORTED. OUR GF-CSL TAKES ResNet50 AS BACKBONE UNLESS SPECIFIED

Methods	Car (07/12)	Plane (07/12)	mAP07	mAP12
YOLOv2 [68]	-/79.20	-/96.60	-	87.90
R-DFPN [69]	-/82.50	-/95.90	-	89.20
S^2ARN [70]	-/92.20	-/97.60	-	94.90
RetinaNet-H [19]	-/93.60	-/97.34	-	95.47
CSL-RetinaNet [23]	88.09/92.93	90.38/97.22	89.23	95.07
BCL-RetinaNet [23]	88.15/92.35	90.57/97.86	89.36	95.01
R3Det [19]	-/94.14	-/98.20	-	96.17
GF-CSL[1]	87.67/90.87	90.42/97.11	89.04	93.99
GF-CSL[1*]	88.28/93.09	90.53/98.35	89.40	95.72
GF-CSL[2]	**88.76**/93.37	90.56/98.11	**89.66**	95.74
GF-CSL[2*]	88.70/**94.15**	90.52/98.69	89.61	96.42
GF-CSL[3†]	88.39/93.05	**90.60**/98.53	89.49	95.79
GF-CSL[3*†]	88.44/93.92	90.58/**99.10**	89.51	**96.51**

[1] The training resolution is 512 pixels.
[2] The training resolution is 640 pixels.
[3] The training resolution is 800 pixels.
[*] The multi-scale testing is used during inference.
[†] The backbone network is ResNet101.

by +0.54% (89.80%) for mAP07 and +0.25% (96.26%) for mAP12 with a lighter backbone (Res50 versus Res101) and a lower training resolution (512 versus 800 pixels). When we only increase the training resolution to 640 pixels, our GF-CSL achieves 90.33% (97.38%) in mAP07 (12). Our best model achieves 97.90% in mAP12, which is a state-of-the-art performance in both one-stage and two-stage methods. Fig. 9 visualizes some detection results from the HRSC2016 dataset.

2) UCAS-AOD: Results on the UCAS-AOD dataset are shown in Table VIII. The cars and PLs in UCAS-AOD usually appear in any direction, and the size of these targets is relatively small; however, experimental results show that our single-stage model achieves the competitive performances of 96.51% for mAP12 and 89.66% for mAP07. The visualization results on the UCAS-AOD dataset are shown in Fig. 10, from which we can see that the cars and PLs can be well detected in most cases.

V. CONCLUSION

In this article, we analyzed the limitations of classification-based rotated object detectors when directly applying existing loss functions to the binary soft labels. Based on that, we proposed the Gaussian FL for angular classification to learn high confident and precise angle predictions. By analyzing the reason why current detectors fail for the square-like objects, we designed an aspect ratio-aware factor based on our Gaussian FL to tackle this problem. The experimental results on the challenging DOTA, HRSC2016, and UCAS-AOD datasets indicated that our proposed GF-CSL could achieve superior performance over the state-of-the-art competitors. In the future, we plan to explore more elegant representations for rotated objects and more efficient detection frameworks in aerial images.

APPENDIX A
SOLUTION TO THE CONTINUOUS FL

In this section, we solve and analyze the optimal solution of continuous FL (2).

When $y = 1$ (Ground-Truth Category): The expression for continuous FL and its derivative with respect to x are as follows:

$$\mathrm{FL}(x) = \alpha(1-x)^{\gamma}\ln\left(\frac{1}{x}\right) \tag{12}$$

$$\frac{\partial \mathrm{FL}(x)}{\partial x} = -\alpha(1-x)^{\gamma-1}\left(\gamma\ln\left(\frac{1}{x}\right)+\frac{1}{x}-1\right) \tag{13}$$

where x is the estimated probability for each angle interval taking values between 0 and 1. $(\partial \mathrm{FL}(x)/\partial x)$ is always negative, and $\mathrm{FL}(x)$ is monotonically decreasing in the range $x \in (0, 1)$. So, the minimum value of $\mathrm{FL}(x)$ is obtained at $x = 1$.

When $y \neq 1$ (Negative Angle Categories):

$$\frac{\partial \mathrm{FL}(x)}{\partial x} = \alpha_t(x+y-2xy)^{\gamma}\left[\frac{1-y}{1-x}-\frac{y}{x}+\frac{\gamma(1-2y)}{x+y-2xy}\left((1-y)\ln\left(\frac{1}{1-x}\right)+y\ln\left(\frac{1}{x}\right)\right)\right] \tag{14}$$

when $x \to 0^+$, $(\partial \mathrm{FL}(x)/\partial x) \to -(\alpha_t y^{\gamma+1}/x) < 0$; when $x \to 1^-$, $(\partial \mathrm{FL}(x)/\partial x) \to (\alpha_t(1-y)^{\gamma+1}/1-x) > 0$. So, $\mathrm{FL}(x)$ first decreases and then increases on the (0, 1) interval. In particular, when $x = y$

$$\left.\frac{\partial \mathrm{FL}(x)}{\partial x}\right|_{x=y} = (1-2y)\cdot\gamma\,\alpha_t\left(2y-2y^2\right)^{\gamma-1}\cdot\left((1-y)\ln\left(\frac{1}{1-y}\right)+y\ln\left(\frac{1}{y}\right)\right) \tag{15}$$

if $y > 0.5$, $(\partial \mathrm{FL}(x)/\partial x)_{x=y} < 0$, the minimum value of $\mathrm{FL}(x)$ is obtained in $(y, 1)$; if $y < 0.5$, $(\partial \mathrm{FL}(x)/\partial x)_{x=y} > 0$, the minimum value of $\mathrm{FL}(x)$ is obtained in $(0, y)$. From this, we can see that when using continuous FL as the loss function, a constant related to the soft label value y needs to be fit at each negative position $(y \neq 1)$.

APPENDIX B
SOLUTION TO THE GAUSSIAN FL

In this section, we solve and analyze the optimal solution of Gaussian FL (7).

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When $y = 1$ (Ground-Truth Category): The Gaussian FL are as follows:

$$\mathrm{GF}(x) = (1-x)^{\gamma} \ln\left(\frac{1}{x}\right) \tag{16}$$

which is similar as (12). So, the minimum value of $\mathrm{GF}(x)$ is also obtained at $x = 1$ in this case.

When $y \neq 1$ (Negative Angle Categories): The Gaussian FL and corresponding derivative are as follows:

$$\mathrm{GF}(x) = (1-y)^{\beta} x^{\gamma} \ln\left(\frac{1}{1-x}\right) \tag{17}$$

$$\frac{\partial \mathrm{GF}(x)}{\partial x} = (1-y)^{\beta} x^{\gamma} \left(\frac{\gamma}{x} \ln\left(\frac{1}{1-x}\right) + \frac{1}{1-x}\right). \tag{18}$$

Note that $(1-y)^{\beta}$ is a constant term that is always greater than zero. $(\partial \mathrm{GF}(x)/\partial x) > 0$ when $x \in (0, 1)$. The minimum value of $\mathrm{GF}(x)$ is obtained at $x = 0$. In contrast to FL, GF does not need to fit intermediate values at non-target angle categories and has and only has $x = 0$ as the optimal solution at these locations.

When $y = y_{\mathrm{sym}}$, where $(\cdot)_{\mathrm{sym}}$ denotes the symmetrical angle position for an ideal square-like object (aspect ratio $= 1$): this is a special case for negative angle categories. When aspect-ratio-aware factor is applied, $y_{\mathrm{sym}} = 1$ for the object. So, no matter what the value of x is, $\mathrm{GF}(x) = (1 - y_{\mathrm{sym}})^{\beta} x^{\gamma} \ln((1/1-x))$ is always equal to 0 because of the Gaussian weight part $(1 - y_{\mathrm{sym}})^{\beta}$. Therefore, the angle prediction result ($x = 1$ when $y = \{y_{\mathrm{true}}, y_{\mathrm{sym}}\}$) for the ideal square-like objects is also the optimal solution of GF.

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